

Problem Set for 13 March 2025

Exercise 1 Let $f : U \rightarrow V$ be linear map and $\dim U = n, \dim V = m$.

1. Let $E \subset V$ be an k -dimensional subspace, prove that

$$\dim f^{-1}(E) \geq n - m + k.$$

答: 考虑映射

$$f_E : f^{-1}(E) \rightarrow V, \quad u \mapsto f(u)$$

此时

$$\begin{aligned} \dim f^{-1}(E) & \stackrel{\text{rank-null}}{=} \dim \ker f_E + \dim \operatorname{im} f_E \\ & \stackrel{\ker f = \ker f_E}{=} \dim \ker f + \dim \operatorname{im} f_E \\ & \stackrel{\text{rank-null}}{=} (\dim U - \dim \operatorname{im} f) + \dim \operatorname{im} f_E \\ & \stackrel{f_E(f^{-1}E)=E}{=} (\dim U - \dim \operatorname{im} f) + \dim E \\ & \geq (\dim U - \dim V) + \dim E. \end{aligned}$$

取等当且仅当 $\dim \operatorname{im} f = \dim V$, 也就是 f 满.

2. When f is surjective, the equality holds.

见上.